

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology & Engineering

University Examination November 2010

CE-204 Digital Electronics and Logic Designs

Date: 29/11/10

Time: 10.30a.m. To 1.30p.m.

Maximum Marks: 70

SECTION-I

Q-1 Attempt the following.

- 1 Add $(1230)_4$ and $(23)_4$ in the given base without converting it to decimal. 01
- 2 Obtain 9's and 10's complement of $(90090)_{10}$. 01
- 3 $(11010111.110)_2 = ()_{10} = ()_{16} = ()_8$. 01
- 4 Convert decimal $(225.225)_{10}$ to binary, octal and hexadecimal. 01
- 5 Using 2's complement perform $(11010)_2 - (1101)_2$ 01
- 6 Add and multiply following numbers in the given base without converting to decimal: $(367)_8$ and $(715)_8$. 01
- 7 Represent decimal number 8620 in Excess-3 code and 2 4 2 1 code. 01

Q-2(a) Design Full Adder with two half adder and an OR gate. 04

(b) Draw and Explain 4-bit binary ripple counter with JK flipflop. 04

(c) Reduce the following Boolean expressions to the required number of literals. 06

$$F = ABC + A'B'C + A'BC + ABC' + A'B'C' \quad (\text{To 5 literals})$$

$$F = BC + AC' + AB + BCD \quad (\text{To 4 literals})$$

OR

Q-2 (a) Design a combinational circuit that multiplies by 5 an input decimal digit represented in BCD. The output is also in BCD. Show that the outputs can be obtained from the input line without using any logic gates. 06

(b) Express the following function in a sum of minterms and product of maxterms. 04

$$F(A, B, C, D) = D(A' + B) + B'D$$

(c) Explain the operation of JK flipflop along with its symbol, characteristic table and characteristic equation. 04

Q-3(a) Determine the prime implicants of the function 06

$$F(w, x, y, z) = \sum m(1, 7, 8, 9, 10, 15).$$

(b) Write a short note on Johnson counter. 05

Given the Boolean function: $F = xy + x'y' + y'z$ 03

(c) (a) Implement it with only OR and NOT gates.

(b) Implement it with only AND and NOT gates.

OR

Q-3 (a) The content of 4 bit shift register is initially 1101. The register is shifted 4 times to right; with the serial input 101101. What is the content of the register after each shift? 02



FOR REFERENCE

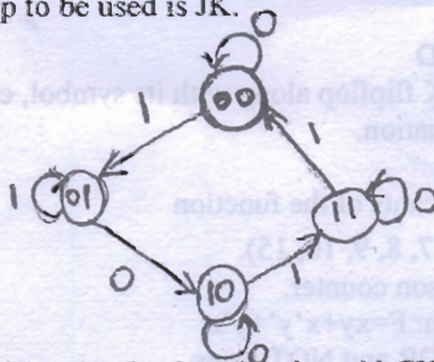
- (b) Write a short note on Look ahead carry generator. 06
- (c) Built the following function in SOP and POS form. 06
- $$f(A,B,C,D) = \sum(0,5,6,11,14) + d(1,2,8,10,15)$$

SECTION II

- Q-4** Do as directed.
- 1 What is the function of Multiplexer? Implement given function with 4x1 Multiplexer: $F(A,B,C) = \sum(1,3,5,6)$ 02
 - 2 What is the disadvantage of RS flipflop? How to remove it? 01
 - 3 Draw the logic diagram of 2 to 4 line decoder with enable(E) input 01
 - 4 How many don't care inputs are there in BCD Adder. 01
 - 5 Define the following terms: Fan in, Fan out, Propagation delay, Power dissipation 02
- Q-5 (a)** Design a counter with the sequence 0,1,3,2,6,4,5,7 and repeat. Use RS flip-flop. 05
- (b) Simplify the following Boolean function using karnaugh map: 04
- $$F(A, B, C, D) = ABD + A'C'D' + A'B + A'CD' + AB'D$$
- (c) Design a BCD to Excess-3 code converter with a BCD to decimal decoder and four OR gates. 05

OR

- Q-5 (a)** A combinational circuit is defined by the functions: 05
- $$F_1(A,B,C) = \sum(3,5,6,7)$$
- $$F_2(A,B,C) = \sum(0,2,4,7)$$
- Implement the circuit with a PLA having three inputs, four product terms and two outputs.
- (b) Design a clocked sequential circuit whose state diagram is given below. The type of flipflop to be used is JK. 07



- (c) Implement the following Boolean function with EX OR and AND gates 02
- $$F = AB'CD' + A'BCD' + AB'C'D + A'BC'D$$
- Q-6(a)** Design a combinational circuit to check for odd parity of 4 bits. 05

- (b) Reduce the number of states in the following state table and tabulate the reduced state table. 07

Present state	Next state		Output	
	X=0	X=1	X=0	X=1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

Starting from the state a of the state table find the output sequence generated with an input sequence 01110010011.

- (c) Define hazards in combinational circuit with their types. 02

OR

- Q-6(a) Simplify the following Boolean function using K-map. 04

$$F(w,x,y,z) = \sum(0,2,4,6,9,11,13,15,17,21,25,27,29,31)$$

- (b) Explain 4 bit binary up down counter with T flipflop. 04

- (c) (a) Design 4x16 decoder with two 3x8 decoder. 06

(b) Design the following Boolean function with NAND gates only. (Use Universal gates concepts)

$$F = A(B+CD) + BC'$$